LEARNMATE – AI STUDYBUDDY

2. Requirement Analysis Phase

Defining the functional, security, technology and infrastructure requirements.

Date	21-09-26
Team ID	SWTID-2026-5714
Project Name	LearnMate - AI StudyBuddy
Maximum marks	4 Marks

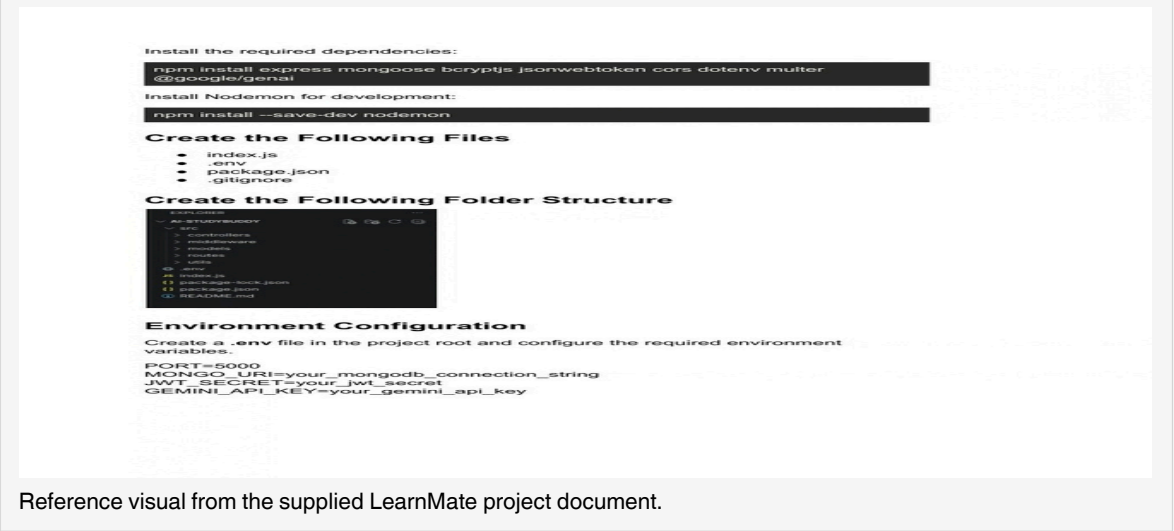


AI-powered learning assistance and smart study support platform

Functional Requirements

Key Points

- User registration and login before accessing AI-powered learning features.
- Upload or enter study materials and select the required AI function.
- Generate summaries, flashcards, quizzes and personalized study plans.



Reference visual from the supplied LearnMate project document.

- Store generated learning resources for future retrieval.
- Provide administrator functions for user and system monitoring.

Security Requirements

Key Points

- JWT-based authentication for protected resources.
- Role-Based Access Control for Student and Administrator privileges.
- Secure password storage using bcryptjs.
- Protected API routes for authenticated users.

Software & Hardware Requirements

Key Points

- Windows 10/11, macOS or Linux; Node.js v16 or above; npm v8 or above.
- Express.js, MongoDB, Postman and Visual Studio Code or similar IDE.
- Intel Core i5 8th Gen or above / AMD Ryzen 5 equivalent; 8 GB RAM minimum; 1 GB available storage.